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Software Test Specification (STS)

Based on IEEE 829 – Adapted for Student Projects

Project Title: InternSync – Cooperative Training Coordination and Tracking System

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Revision History

Version	Date	Description	Author
1.0	Nov 1, 2026	First version for Project 7 (STS) on the course template (IEEE 829 adapted), from the acceptance tests and test reports of Increments 1 to 5, the UAT plan, and the key results.	Group 3

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1. Introduction

1.1 Purpose

This Software Test Specification (STS) defines how InternSync is tested: what is tested and what is not, at which levels and with which tools, the test cases with their inputs, expected and actual results, the test data and environment, who is responsible for each part, and when it runs. It follows IEEE Std 829 as adapted in the CSC 305 template, and it records the results of the test runs of Increments 1 to 5.

1.2 Scope of Testing

Level	What it covers	How
Unit	Single functions: deadlines, state transitions, grade computation, skill extraction, scoring, passage splitting, date format, translation keys	pytest and Vitest, automated
Integration (API)	Each endpoint with the real PostgreSQL database: business rules, permissions, scope across organizations, audit, notifications, files	pytest with the HTTP test client, automated on every pull request
System	The whole stack in a browser: role tasks, both languages, offline mode, phones and desktops; plus load, security, and AI accuracy runs	Playwright, Locust, OWASP ZAP, the AI evaluation script
Acceptance	One acceptance test per requirement (Appendix 11.2), and user acceptance testing with at least 8 participants	Automated where possible; UAT sessions with scenario cards

Every requirement of the SRS has an acceptance test or a recorded check. The tests run against the system as merged on `main` at the end of Increment 5.

1.3 Reference Documents

Document	Use in this STS
SRS 1.6	Requirements and their acceptance criteria
SDD 1.0 and the System Design Document 1.7	State machines, algorithms, thresholds, and the test design
SPMP 1.7	Roles, schedule, and milestones M3 – M7
Increment documents (1, 2, 3, 4)	Acceptance tests of each increment (Section 6 of each)
Test reports (1, 2, 3, 4, 5)	Execution results and defect logs
UAT plan	Participants, scenario cards, and the result sheet
Key results and Requirements status	Measured results and requirement-to-test tracing
IEEE Std 829-1998	IEEE Standard for Software Test Documentation

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2. Test Items

The item under test is InternSync as merged on `main` at the end of Increment 5 (commit `0662647`): the backend image, the web app build, and the database migrations.

Item	Parts	Main tests
Authentication and access control	Registration, login, sessions, password reset, lockout, consent, role and scope gates, permission matrix	`backend/tests/test_auth.py`, `tests/permissions`
Organizations and verification	Universities, colleges, companies, verification requests and documents, invitations, role transfer	`test_orgs.py`
Training cycles	Weeks and deadlines, grade components, rubric, enrolments, closing and reopening	`test_cycles.py`
Student profile and CV	Profile, CV upload and type checks, skill extraction and confirmation	`test_profiles.py`
Vacancies and placement	Vacancies, applications, placements, filters	`test_vacancies.py`
Matching (AI)	Eligibility, fit score, explanation, development plan, basic fallback	`test_matching.py`, `ai-eval/evaluate.py`
Reports	Report slots, drafts, attachments, submission, review, comments, lateness, missing flag	`test_reports.py`
Similarity detection (AI)	Passages, checks, flags, decisions	`test_similarity.py`, `ai-eval/evaluate.py`
Attendance, evaluations, grades	Attendance and unlocks, evaluations, grade computation and approval workflow	`test_attendance_evaluations.py`, `test_grades_dashboard.py`
Dashboards, exports, certificates	Pending actions, statistics, leniency alert, grade sheet, term summary, training record, certificates	`test_grades_dashboard.py`, `test_feedback.py`, `test_exports.py`, `test_certificates.py`
Notifications and scheduler	Outbox, email, web push, reminders, missing flags, escalations, job locks	`test_notifications.py`, `test_audit_jobs.py`
Audit and privacy	Audit log, history, anonymization, purge of verification documents	`test_audit_jobs.py`, `test_privacy.py`
Web app (PWA)	Screens of every role, Arabic and English, offline mode, install, accessibility, layouts	`frontend/src/**/*.test.ts`, `frontend/e2e/*.spec.ts`
Database and storage	Migrations, constraints, row level security, append-only audit trigger, file storage adapters	Every backend test on a fresh database; `test_storage.py`, `test_config.py`
Demo scenarios	The M3 to M6 demonstration scripts end to end through the API	`test_demo_scenario.py`, `test_demo_m4.py`, `test_demo_m5.py`, `test_demo_m6.py`

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3. Features to Be Tested

Every feature below is traced from its SRS requirements to the design and to its acceptance tests (Appendix 11.2). The requirements status lists all 125 requirements with their tests.

Feature	SRS requirements	SDD section	Acceptance tests
Login, sessions, and roles	FR-AUT-01 – 06, NFR-SEC-02/03	5.8	AT-AUT-01 – 06
Registration and verification of organizations and students	FR-ORG-01 – 09	5.1, 5.3	AT-ORG-01 – 09
Training cycles and configuration	FR-CYC-01 – 05	5.6.5	AT-CYC-01 – 05
Student profile and CV skill extraction	FR-PRF-02 – 05	5.6.1	AT-PRF-02 – 05
Vacancies, applications, and placements	FR-VAC-01 – 08	5.3	AT-VAC-01 – 08
Matching, explanation, and development plan	FR-MAT-01 – 07, NFR-AI-02/03	5.6.2	AT-MAT-01 – 07, AT-AI-02/03
Weekly and final reports, and their workflow	FR-REP-01 – 07, FR-WFL-01	5.2, 5.3	AT-REP-01 – 07, AT-WFL-01
Attendance	FR-ATT-01 – 04	5.1	AT-ATT-01 – 04
Evaluations	FR-EVL-01 – 05	5.1	AT-EVL-01 – 05
Grades and approval	FR-WFL-02 – 07	5.3, 5.6.4	AT-WFL-02 – 07
Similarity detection	FR-SIM-01 – 03	5.6.3	AT-SIM-01 – 03
Notifications and reminders	FR-NOT-01 – 06, NFR-REL-03	5.7	AT-NOT-01 – 06, AT-REL-03
Dashboards and statistics	FR-DSH-01 – 05	5.1	AT-DSH-01 – 05
Exports and completion certificates	FR-EXP-01 – 03, FR-CER-01 – 03, NFR-L10N-03	5.1	AT-EXP-01 – 03, AT-CER-01 – 03, AT-L10N-03
Language and installable app	FR-LOC-01/02, FR-PWA-01 – 03	5.9	AT-LOC-01 – 03, AT-PWA-01 – 03
Audit trail and privacy	FR-AUD-01 – 03, NFR-PRV-01 – 04	5.8, 6.4	AT-AUD-01 – 03, AT-PRV-04
Performance	NFR-PER-01 – 05	4.4	AT-PER-01 – 05
Security, portability, and usability	NFR-SEC-04/05, NFR-PRT-01, NFR-USA-03/04	5.8	AT-SEC-05, AT-PRT-01, AT-USA-04

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4. Features Not to Be Tested

Item	Reason
Services of other providers: Gmail delivery, the browser vendors' push services, Vercel, Hugging Face, Supabase	Outside the project; only the system's calls to them are tested (emails caught by Mailpit, push sent through a stand-in sender)
Uptime of the hosted system (NFR-REL-01) and HTTPS on it (NFR-SEC-01)	The reference deployment is prepared but switched on only by the owner; checked after the switch-on (Section 9.2)
Load above 50 concurrent users	The SRS asks for 50 users (NFR-PER-01); about 400 students in the demonstration
Real (non-synthetic) personal data	Only synthetic data is used (SRS C-09); no real CV, report, or grade enters the system
Old browsers and Internet Explorer	Outside the supported environment (SRS 2.4); current Chromium, WebKit, and Firefox are tested
Native mobile apps	There are none; the web app is installable on phones as a PWA
Framework messages for wrongly typed values (for example letters in a number field)	They come from the framework in English; the forms prevent most of them, and every business rule answers in the user's language
Wireframes	Design artefacts, reviewed in task T3, not executable

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5. Test Approach

5.1 Strategy

- **Automated first.** Every acceptance test is automated where possible and named after its ID (for example `test_at_rep_04_on_time_late_and_missing`), so each result traces to a requirement. Manual checks are kept for what needs eyes: layouts in both languages, the shaped Arabic in PDFs, and installation on a phone.
- **Black-box at the API.** Most tests call the HTTP API as a real user of a given role and check status codes, error codes, stored data, audit entries, and notifications. White-box unit tests cover the algorithms (deadlines, grade total, matching score, passage splitting) with exact expected values.
- **Positive and negative cases.** Each rule is tested where it allows and where it refuses: every allowed state transition passes, and at least one disallowed transition from each state is rejected.
- **Permissions exhaustively.** A matrix lists every endpoint with the callers allowed past the role gates; one test calls each endpoint as every other caller and expects `401` or `403`, and another fails if an endpoint is missing from the matrix. Scope across organizations is tested with two universities, three colleges, and two companies.
- **Controlled time.** The application reads time from one clock that tests can move, so deadlines, the 30-minute idle limit, the 15-minute lockout, purges after 30 days, and escalations after 3 days are tested without waiting.
- **Honest AI numbers.** The AI is tuned on labelled tuning sets, then measured once on held-out sets written after the settings were frozen.
- **Regression on every change.** CI runs lint, the backend tests with a 70% coverage gate, the API contract check, the frontend tests and build, the dependency audits, and the browser tests on every pull request.

5.2 Tools

Tool	Used for
pytest, pytest-cov, httpx test client	Unit and API tests on a real PostgreSQL database; coverage
Vitest	Frontend unit tests (translation keys, date format, offline storage)
Playwright	Browser tests on Chromium, WebKit, Firefox, and phone sizes; offline mode; keyboard-only task
axe-core (through Playwright)	Accessibility: colour contrast and keyboard use
Locust	Load test with 50 concurrent users
OWASP ZAP	Baseline scan of the web app and API scan in safe mode
pip-audit, npm audit	Known vulnerabilities in dependencies
`ai-eval/evaluate.py`	Skill extraction recall, matching top-5 rate, similarity detection rates
Mailpit	Catches every email the system sends
GitHub Actions	Runs the suites on every pull request

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5.3 Entry and Exit Criteria

Criterion	Rule
Entry	The increment's acceptance tests are written in its requirements document; the build passes lint and compiles
Pass of a test case	The actual result equals the expected result, including the status and error code
Exit of an increment	Every acceptance test passes or has a recorded reason; coverage ≥ 70%; no open defect of high severity; the demo dry run passes
Defects	Logged in the test report of the increment with how it was found and the fix; the fix gets a regression test
Suspension	A failing migration or a broken build stops testing until fixed

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6. Test Cases

6.1 Functional Test Cases

Each case comes from an acceptance test of Appendix 11.2 (named in the Feature column). Positive cases check that a valid action succeeds; negative cases check that an invalid one is refused.

Table 6.1 – Functional test cases

Test Case ID	Feature	Input	Expected Output	Actual Result	Pass/ Fail
TC-01	Login by role (AT-AUT-01)	Each demo account (student, company supervisor, Company Admin, academic supervisor, coordinator, Platform Admin) signs in with its password	Each role gets only its own menu; the Company Admin also gets the team menu	Each role saw only its menu; team menu for the Company Admin only	Pass
TC-02	Account lockout (NFR-SEC-03)	Five wrong passwords for one account, then the right password; the clock moved 16 minutes	Locked after the fifth failure (`423 ACCOUNT_LOCKED`), even with the right password; open again after 15 minutes	Locked, then open after 16 minutes	Pass
TC-03	Idle timeout and logout (AT-AUT-03)	A request after 30 idle minutes; the old token after logout	`401` in both cases	`401 SESSION_EXPIRED` in both cases	Pass
TC-04	One-time links (AT-AUT-02, AT-AUT-04)	An email confirmation link used twice; an expired link; a password reset link used twice	The first use works; a second or expired use is rejected; the old password fails after the reset	As expected	Pass
TC-05	Permission matrix (AT-AUT-05)	All 163 endpoints called by every caller not allowed (public, unverified, ST, CS, CA, AS, CO, PA)	`401` or `403` for every call	810 refused calls, none got through	Pass
TC-06	Scope across colleges (AT-ORG-05)	The coordinator of college B opens the verification request of a student of college A	`403 FORBIDDEN`	`403`	Pass
TC-07	Student verification (AT-ORG-05)	The coordinator approves a student's request and chooses the cycle	The student is verified and enrolled and can apply	Verified, enrolled, application accepted	Pass
TC-08	Rejection needs a reason (AT-ORG-03, AT-ORG-05)	A coordinator or the Platform Admin rejects a request with an empty reason	Refused with `422`	`422`	Pass
TC-09	Unverified accounts (AT-ORG-02)	An unverified student applies; an unverified company creates a vacancy	`403 VERIFICATION_REQUIRED`	`403 VERIFICATION_REQUIRED`	Pass
TC-10	Duplicate company (AT-ORG-04)	A company registers with a commercial registration number already used	Refused with an error	Refused	Pass
TC-11	Weekly deadlines (AT-CYC-01)	A 12-week cycle with report day Thursday at 23:59	12 deadlines, each on Thursday at 23:59 Riyadh time; scheduled days equal the working days	12 deadlines on the right day and time	Pass
TC-12	Grade weights (AT-CYC-02)	Weights 40/30/20, then 40/30/30	40/30/20 refused; 40/30/30 saved	As expected	Pass
TC-13	CV upload checks (AT-PRF-03, NFR-SEC-04)	A 6 MB PDF; a PNG renamed `.pdf`; a ZIP renamed `.docx`; a second valid CV	Oversized file `413`; wrong types `415`; both valid versions kept	As expected	Pass

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Test Case ID	Feature	Input	Expected Output	Actual Result	Pass/ Fail
TC-14	Skill extraction (AT-PRF-04, AT-CYC-04)	An English PDF CV, an Arabic DOCX CV, and a CV containing "JS"	Non-empty candidate lists; "JS" shown as "JavaScript"; only confirmed skills become profile skills	As expected	Pass
TC-15	CV without text (AT-PRF-05)	An image-only PDF	The "no readable text" message and an empty editable skill list	Message and empty list shown	Pass
TC-16	Vacancy visibility (AT-VAC-02)	A draft vacancy; a vacancy open only to university A	The draft is hidden from students; the targeted vacancy is visible only to A's students	As expected	Pass
TC-17	Application rules (AT-VAC-04)	A duplicate application, one after the deadline, one of the wrong training type, and a sixth active one	Each refused with its own message	Each refused with its message	Pass
TC-18	Placement (AT-VAC-07)	The coordinator confirms a placement on the last seat, then a second placement for the same student	Placement confirmed; vacancy closed; the student's other applications withdrawn; second placement refused	As expected	Pass
TC-19	Ranked vacancies (AT-MAT-01)	A student profile with confirmed skills; seeded vacancies	Only eligible vacancies, by fit score from high to low; a relevant vacancy in the top 5 for at least 80% of labelled profiles	Ordered by fit; relevant vacancy in the top 5 for 100%	Pass
TC-20	Ineligible vacancy (AT-MAT-03)	A vacancy whose minimum GPA is above the student's GPA	Absent from the ranking; shows "GPA below minimum" when browsing	As expected	Pass
TC-21	Basic matching (AT-MAT-04)	Ranking with the model switched off (`AI_ENABLED=false`)	A ranking within 5 s labelled "Basic matching"	Returned and labelled basic	Pass
TC-22	Match explanation (AT-MAT-05)	A student and a vacancy with some required skills missing	Matched skills = confirmed ∩ required; missing skills = required − confirmed	Sets equal	Pass
TC-23	Report slots (AT-REP-01)	A placement in a 12-week cycle	12 weekly slots and 1 final slot with the right deadlines	As expected	Pass
TC-24	Report attachments (AT-REP-02)	A fourth attachment; an 11 MB file; a valid report with 3 files	Fourth file and 11 MB file refused; the valid report stored with its files	As expected	Pass
TC-25	Required fields (AT-REP-02)	A weekly report submitted with "challenges" empty	`422 REQUIRED_FIELDS_MISSING` naming the field	`422` with the field	Pass
TC-26	On time, late, and missing (AT-REP-04)	Deadline 23:59: submissions at 23:58 and 00:00; a slot left empty	23:58 On time; 00:00 Late; the empty slot flagged Missing within 1 hour	On time, Late; Missing at the next job run (every 15 minutes)	Pass
TC-27	Revision cycle (AT-REP-05)	A revision request with no comment; a revision with a comment, then a resubmission	No comment refused; after resubmission both versions viewable	As expected	Pass
TC-28	Invalid transitions (AT-WFL-01)	Accept a draft report; submit to a slot closed as missing	`409 INVALID_TRANSITION`; `409 SLOT_CLOSED`	As expected	Pass
TC-29	Attendance (AT-ATT-01, AT-ATT-02)	"Full attendance" for a week; more days than scheduled; an edit on day 8 after the week, before and after an unlock	Full week saved in one action; extra days refused; edit refused until the coordinator unlocks the week (audited)	As expected	Pass

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Test Case ID	Feature	Input	Expected Output	Actual Result	Pass/ Fail
TC-30	Evaluations (AT-EVL-01, AT-EVL-04)	An evaluation with an unrated criterion or no comment; a second reopening by the coordinator	Both refused; the first reopening needs a reason and is audited	As expected	Pass
TC-31	Grade computation (AT-WFL-02)	Seeded component scores and weights	The total equals a hand calculation to two decimals	Equal	Pass
TC-32	Grade preconditions (AT-WFL-03)	The academic supervisor submits a grade while the final report is not accepted	`409 PRECONDITIONS_UNMET` with the list of missing items	Refused with the list	Pass
TC-33	Grade approval (AT-WFL-04, AT-WFL-06)	The coordinator approves a submitted grade; the student views the grade before and after	Final with a timestamp; placement completed; the student sees the grade only when Final	As expected	Pass
TC-34	Similarity flag (AT-SIM-01, AT-SIM-02)	A report copied from an earlier report of the same university; the same text stored at another university	Flagged with the passages side by side; reports of another university never compared	As expected	Pass
TC-35	Similarity decision (AT-SIM-03)	A decision without a comment; a decision on a check that is not flagged	Both refused; a valid decision changes neither the report status nor the grade	As expected	Pass
TC-36	Notifications (AT-NOT-01, AT-NOT-06)	A report is submitted; the supervisor has chosen Arabic	One in-app notification and one email, in Arabic; the unread count drops when read	As expected	Pass
TC-37	Reminders and escalations (AT-NOT-02, AT-NOT-04)	Unsubmitted slots due within 24 hours and later; items 3 and 4 days overdue; each job run twice	Reminders only for the slots due within 24 hours, once each; escalation on day 4, not day 3, never twice	As expected	Pass
TC-38	Grade sheet export (AT-EXP-01)	A cycle with students whose names are in Arabic	Excel and PDF values equal the stored values; Arabic names present and shaped	As expected	Pass
TC-39	Certificate (AT-CER-01, AT-CER-02)	Issue for a grade that is not Final; issue for a Final grade; open the public page with the code and with an altered code	First refused; certificate with a 26-character code and QR code; page shows Valid without the grade; altered code shows Not found	As expected	Pass
TC-40	Offline mode (AT-PWA-03)	The network turned off after opening pages; a report draft typed offline; the network back	Saved copy shown with its time; a never-opened page says it is not saved; submit disabled; the draft saved when back online	As expected in Chromium and WebKit	Pass
TC-41	Language (AT-LOC-01, AT-LOC-02)	Switch each screen between Arabic and English; run the translation-key check	Screens mirror right to left and left to right; the choice is saved; no missing key	839 texts in both languages; no missing key	Pass
TC-42	Audit log (AT-AUD-01, AT-AUD-02)	Each action of the demo script; an `UPDATE` and a `DELETE` on `audit_log` in the database	Exactly one audit entry per action; the database refuses to change entries	As expected	Pass
TC-43	Anonymization (AT-PRV-04)	Anonymize a closed cycle before and after its retention period	Refused before; after it no name, email, student number, CV, report text, or certificate file can be retrieved, and the statistics stay	As expected	Pass

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6.2 Non-Functional Test Cases

Table 6.2 – Non-functional test cases

Test Case ID	Feature	Input	Expected Output	Actual Result	Pass/ Fail
NF-01	Response time under load (AT-PER-01)	50 simulated users for 5 minutes (Locust), local stack	95% of ordinary requests within 2 s	95% within 0.21 s; 0 errors in 7,785 requests	Pass (local stack)
NF-02	Ranking speed (AT-PER-02)	1,000 published vacancies; 20 ranking requests per mode	95th percentile ≤ 5 s, semantic and basic	0.83 s semantic, 0.76 s basic	Pass
NF-03	CV processing time (NFR-PER-03)	A 4.8 MB PDF packed with text	Skills suggested within 30 s	About 2–6 s	Pass
NF-04	Similarity at scale (AT-PER-04)	A new report checked against 10,000 stored reports	Result within 2 minutes	0.35–0.42 s	Pass
NF-05	Export speed (AT-PER-05)	Grade sheet of 100 students	Excel and PDF each under 60 s	Excel 1.0–1.3 s, PDF 2.4–4.6 s	Pass
NF-06	Job idempotency (AT-REL-03)	Each scheduled job run twice; a simulated downtime	Nothing sent twice; the next run catches up	As expected	Pass
NF-07	Security scan (AT-SEC-05)	ZAP baseline and API scan; pip-audit; npm audit	No high-risk finding; no critical vulnerable dependency	0 high-risk after two fixes; 0 vulnerable dependencies	Pass
NF-08	Browsers and screen sizes (AT-PRT-01)	Browser tests on Chromium, WebKit, Firefox, and widths 360, 768, 1366, 1920 px	All pass with no horizontal scrolling	34 passed on Chromium, WebKit, Pixel 7, iPhone 14; Firefox in CI	Pass
NF-09	Accessibility (AT-USA-04)	axe check of the main pages; a main task with the keyboard only	No contrast failure; task completed by keyboard	One contrast failure (3.86:1) fixed, then none	Pass
NF-10	Arabic in PDFs (AT-L10N-03)	Exported PDF with Arabic names	The Arabic font embedded; text joined and right to left	As expected	Pass
NF-11	Skill extraction accuracy (FR-PRF-04)	20 labelled CVs, Arabic and English	Recall ≥ 70%	97.2% (English 97.4%, Arabic 96.9%)	Pass
NF-12	AI on held-out sets (AT-AI-03)	Held-out sets run once with the frozen settings	Relevant vacancy in the top 5 for ≥ 80% of profiles; ≥ 9 of 10 copied reports flagged; ≤ 1 of 10 originals flagged	Matching 100%; 7 of 10 copied flagged (the 3 reworded ones missed); 0 of 10 originals flagged	Fail (similarity part)
NF-13	Coverage (NFR-MNT-01)	`pytest --cov=app`	≥ 70% of the backend code	95%	Pass

NF-12 is the one recorded failure: the similarity check finds text copied word for word, not reports reworded sentence by sentence. It stays advisory, a supervisor decides, and the limit is stated in the final report.

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6.3 Execution Summary

Run	Date	Acceptance tests	Other results
Increment 1	Oct 24, 2026	43 of 43 passed	108 backend tests; 494 refused calls in the permission matrix
Increment 2	Oct 26, 2026	27 of 27 passed	158 backend tests; matching on the tuning set
Increment 3	Oct 27, 2026	21 of 21 passed in their automated part	187 backend tests; device parts of AT-PWA-01 – 03 moved to M6
Increment 4	Oct 28, 2026	12 of 13 met (AT-AI-03 not met for similarity)	201 backend tests; 34 browser tests; ZAP, load test, held-out AI sets
Increment 5	Oct 29, 2026	Regression	202 backend tests, 19 frontend tests, 34 browser tests, all passing
User acceptance testing	Planned for M6	Scenario cards for the five roles	Postponed on Oct 29; open (Section 9.2)

102 distinct acceptance tests were run 104 times (AT-AI-03 in Increments 2 and 4, AT-PWA-03 in Increments 3 and 4); 101 are met and one is met in part. 35 defects were found and fixed (Section 10).

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7. Test Data

Dataset	Content	Valid and invalid cases
Test fixtures (`backend/tests`)	Two universities, three colleges, two companies, users of every role, cycles, vacancies, placements; a fresh database per test run	Records of another college or company for the scope tests
Demo data (`python -m app.seed --demo`)	One account per role (password `Demo-2026-pass`, refused in production), students in every state (placed, accepted and waiting, not applied, former students), a closed summer 2025 cycle with Final grades	Late and missing reports, a copied report, a generous company for the leniency alert
Files	Generated PDF and DOCX CVs in Arabic and English, an image-only PDF, a 4.8 MB PDF, report attachments	A 6 MB CV, an 11 MB attachment, a PNG renamed `.pdf`, a ZIP renamed `.docx`, plain text
AI tuning sets (`ai-eval`)	20 labelled CVs, profiles with their relevant vacancies, 10 copied and 10 original reports	Copied reports at 44–54% copied text; originals on the same topics
AI held-out sets (`ai-eval --heldout`)	New profiles with 15 eligible vacancies each; 10 copied and 10 original reports, written after the settings were frozen	3 reports reworded sentence by sentence
Load and scale data	1,000 published vacancies; 10,000 stored reports; 100 students for the grade sheet; 50 Locust users	–
Invalid inputs	Weights 40/30/20, rejection without a reason, expired and used links, a sixth application, wrong training type, edits after a deadline, disallowed transitions from every state	Each must be refused with its error code

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8. Test Environment

Part	Configuration
Development laptops	Windows with Docker Desktop; the whole stack with `docker compose up` (database, API, web app, Mailpit)
Backend	Python 3.11, FastAPI; PostgreSQL 16 with pgvector (`pgvector/pgvector:pg16`) and `pg_trgm`; a separate test database `internsync_test`
Frontend	Node 22, React 18, Vite, TypeScript strict
Browsers	Chromium and WebKit (Playwright), Firefox in CI; phone sizes Pixel 7 and iPhone 14; widths 360, 768, 1366, and 1920 px
CI	GitHub Actions on Ubuntu: lint, backend tests with a PostgreSQL service container, OpenAPI and client checks, frontend tests and build, dependency audits, and browser tests against the Docker stack on its own database (`docker-compose.check.yml`)
Email	Mailpit catches every email (inbox on port 8025)
AI model	`intfloat/multilingual-e5-small` on CPU; a small stand-in model in the unit tests
Release checks	Locust, OWASP ZAP (Docker), pip-audit, npm audit
Hosted system (after switch-on)	Vercel, a Hugging Face Space, Supabase, Gmail SMTP, for the checks of Section 4

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9. Responsibilities and Schedule

9.1 Responsibilities

Team Member	Testing responsibilities	Test cases
Salam Alsaour (Team Leader)	Test strategy and integration; CI; tests of reports, the report workflow, grades, and similarity; reviews every test pull request	TC-23 – TC-28, TC-31 – TC-35, NF-04, NF-12, NF-13
Khalid Alajmi	Tests of authentication, access control, the permission matrix, organizations and verification, vacancies, applications, and placements	TC-01 – TC-10, TC-16 – TC-18, NF-06
Ali Alghafly	Frontend tests: translation keys, both languages and directions, offline mode, browser tests on all engines and sizes	TC-40, TC-41, NF-08
Ahmed Kanaan	AI tests and evaluation (skill extraction, matching, held-out sets); load test; checks on the hosted system after the switch-on	TC-14, TC-15, TC-19 – TC-22, NF-01 – NF-03, NF-11
Abdullah Abalhareth	QA lead: this STS, the acceptance tests of each increment, cycles, CV checks, attendance, evaluations, notifications, exports, certificates, audit, and privacy; the security scan; the defect log; organizes the UAT	TC-11 – TC-13, TC-29, TC-30, TC-36 – TC-39, TC-42, TC-43, NF-07
Rakan Alzahrani	Usability and accessibility: the accessibility check, the Arabic PDF layout, the UAT scenario cards and sessions, the device check on phones	NF-09, NF-10, the UAT scenarios

9.2 Schedule

Activity	Dates	Status
Acceptance tests written for Increment 1 (with the requirements)	Oct 16 – 23, 2026	Done
Increment 1 test run and M3 dry run	Oct 24, 2026	Done (43 of 43)
Increment 2 test run, AI tuning sets	Oct 26, 2026	Done (27 of 27)
Increment 3 test run, similarity evaluation	Oct 27, 2026	Done (21 of 21)
Increment 4 release checks: browser tests, ZAP, load test, held-out AI sets	Oct 28, 2026	Done (12 of 13)
Increment 5 regression	Oct 29, 2026	Done
Regression on every pull request	Continuous	Running in CI
Milestone demos M3, M4, M5	Nov 1, Nov 15, Nov 25, 2026	Planned
User acceptance testing with at least 8 participants, and the device check on phones	By Dec 10, 2026 (M6)	Postponed on Oct 29; to be scheduled
Checks on the hosted system: uptime, HTTPS, load test repeated	After the switch-on, before Dec 10, 2026	Waiting for the owner's accounts
Final regression before the demonstration	Dec 14 – 15, 2026 (M7)	Planned

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10. Deliverables

Deliverable	Where
Test cases document	This STS; the acceptance tests in Section 6 of each increment's requirements document
Automated test suites	`backend/tests` (202 tests and the permission matrix), `frontend/src/**/*.test.ts` (19 tests), `frontend/e2e` (browser tests), `backend/loadtest`, `ai-eval`
Test execution reports	The test report of each increment (`docs/Increments/Increment_N/04_Test_Report.md`) and the key results
Defect log	Section "Defects Found and Fixed" of each test report: 8, 9, 8, 7, and 3 defects in Increments 1 to 5, each with how it was found and its fix
UAT materials	The UAT plan with the consent text, scenario cards for the five roles, the result sheet, and the device check
Security and load evidence	ZAP reports (kept 30 days as workflow artifacts), the Locust run, the dependency audits in CI
Traceability	The requirements status: 125 requirements, 90 with an automated test named after them, 32 with a recorded result, 3 open

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11. Appendices

11.1 Glossary

Term	Meaning
Acceptance test (AT)	A test that checks one requirement's acceptance criterion; named `AT-<area>-<number>`
Regression testing	Running the existing tests again after a change to make sure nothing that worked is broken
Test stub	A simple stand-in for a part of the system that is not under test, such as the push sender used in the tests
Mock data	Made-up data with the shape of real data; InternSync uses only synthetic data
Fixture	Data and set-up prepared before a test, such as the two universities and two companies
Permission matrix	The table of every endpoint with the callers allowed past its role gates; tested for every other caller
End-to-end (E2E) test	A test that drives the app in a real browser, from the screen to the database
Held-out set	Test data kept aside and used once, after the settings were chosen, so the result is not tuned to it
Coverage	The share of the code run by the tests
Load test	A test with many simulated users at once to measure response times
Idempotent	Running it twice has the same effect as running it once
UAT	User acceptance testing: real users try the main tasks with scenario cards

11.2 Acceptance Tests and Results

The table lists every acceptance test with its requirement, its scenario and expected result, its level, and its latest result. Levels: *Unit* checks one function; *API* calls the HTTP API with a test database; *E2E* drives the browser; *Eval* runs the AI evaluation; *Load* and *Scan* are the release checks.

Table 11.1 – The 102 acceptance tests and their latest results

Test	Requirement	Scenario and expected result	Level	Result
AT-AUT-01	FR-AUT-01	A user of each role logs in and receives only that role's menu; a Company Admin also gets the team menu	API, E2E	Pass
AT-AUT-02	FR-AUT-02	Registration offered only for student, coordinator, company; confirmation and invitation links work once; used or expired link rejected	API	Pass
AT-AUT-03	FR-AUT-03	After 30 idle minutes the next request returns 401; after logout the old token is rejected	API	Pass
AT-AUT-04	FR-AUT-04	Reset link works once; old password fails afterwards; expired link rejected	API	Pass
AT-AUT-05	FR-AUT-05	Permission matrix test: every Increment 1 endpoint × every role, with 2 colleges and 2 companies; 0 failures, no access across organizations	API	Pass
AT-AUT-06	FR-AUT-06	Deactivated user cannot log in; past records still visible; a Coordinator cannot deactivate a user of another college	API	Pass
AT-ORG-01	FR-ORG-01	A new college appears in the registration options; a deactivated college does not	API	Pass

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Test	Requirement	Scenario and expected result	Level	Result
AT-ORG-02	FR-ORG-02	Unverified student cannot apply; unverified coordinator sees no student; unverified company cannot create a vacancy	API	Pass
AT-ORG-03	FR-ORG-03	Second coordinator registration for a college refused; rejection without reason refused	API	Pass
AT-ORG-04	FR-ORG-04	Duplicate commercial registration number refused; before approval no vacancy and no invitation	API	Pass
AT-ORG-05	FR-ORG-05	After approval the student is enrolled and can apply; rejection without reason refused; another college's coordinator cannot see the request	API	Pass
AT-ORG-06	FR-ORG-06	Invited academic supervisor belongs to the inviting coordinator's college; used or expired invitation rejected	API	Pass
AT-ORG-08	FR-ORG-08	Another coordinator gets 403 on the document; after the purge job (clock set 31 days ahead) the file returns 410 and the decision remains	API	Pass
AT-ORG-09	FR-ORG-09 (stretch)	After a rejection a new request is accepted; the reviewer sees the earlier decision	API	Pass
AT-CYC-01	FR-CYC-01	A 12-week cycle generates 12 weekly deadlines on the configured weekday and time; scheduled days equal the working days	Unit, API	Pass
AT-CYC-02	FR-CYC-02	Weights 40/30/20 refused; 40/30/30 saved	API	Pass
AT-CYC-03	FR-CYC-03	With the lock timestamp set, editing the rubric is refused	API	Pass
AT-CYC-04	FR-CYC-04	Skill with synonym "JS" is recognized in a CV containing "JS" and shown as "JavaScript"	Unit, API	Pass
AT-CYC-05	FR-CYC-05 (stretch)	Edit in a closed cycle refused; reopening without reason refused	API	Pass
AT-PRF-02	FR-PRF-02	After verification the student edits skills and interests but not student number, major, or GPA	API	Pass
AT-PRF-03	FR-PRF-03	6 MB file or other type refused; second upload keeps both versions	API	Pass
AT-PRF-04	FR-PRF-04	Sample Arabic and English CVs produce a non-empty candidate list; only confirmed skills are stored as profile skills	Unit, API	Pass
AT-PRF-05	FR-PRF-05	Image-only PDF shows the message and an empty editable list	API, E2E	Pass
AT-VAC-01	FR-VAC-01	Missing required field cannot be saved; saved vacancy shows all fields; unverified company refused	API	Pass
AT-VAC-02	FR-VAC-02	Draft invisible to students; vacancy targeted at university A visible to A's students only	API	Pass
AT-VAC-03	FR-VAC-03	Each filter returns only matching vacancies on seeded data	API	Pass

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Test	Requirement	Scenario and expected result	Level	Result
AT-VAC-04	FR-VAC-04	Duplicate, after-deadline, wrong-training-type (CR-06), and sixth active applications each refused with a message	API	Pass
AT-VAC-05	FR-VAC-05 (stretch)	Withdrawn application leaves the list and frees a slot	API	Pass
AT-VAC-06	FR-VAC-06	Supervisor sees own vacancies' applicants only; Company Admin sees the company's; decision visible to student and coordinator	API	Pass
AT-VAC-07	FR-VAC-07	Second placement for the student refused; filling the last seat closes the vacancy; other active applications withdrawn	API	Pass
AT-VAC-08	FR-VAC-08 (stretch)	Change or cancel without reason refused; change in the audit log	API	Pass
AT-REP-01	FR-REP-01	Placement in a 12-week cycle has 12 weekly slots and 1 final slot with correct deadlines	API	Pass
AT-REP-02	FR-REP-02	Fourth attachment or 11 MB file refused; valid submission stored with its files	API	Pass
AT-REP-03	FR-REP-03	Editing after submission refused; supervisors cannot see drafts	API	Pass
AT-REP-04	FR-REP-04, NFR-L10N-02	Submissions 1 minute before and after the deadline flagged On time and Late; empty slot Missing within 1 hour after the deadline (job run)	Unit, API	Pass
AT-REP-05	FR-REP-05	Revision request without comment refused; after resubmission both versions viewable	API	Pass
AT-REP-06	FR-REP-06	Comment visible to the other party with author and time; company supervisor read-only	API	Pass
AT-REP-07	FR-REP-07	After closing, the student cannot submit to that slot	API	Pass
AT-WFL-01	FR-WFL-01	Every allowed report transition passes; at least one disallowed transition from each state is rejected	Unit	Pass
AT-LOC-01	FR-LOC-01	Switching language mirrors every Increment 1 screen; the choice is saved per user	E2E	Pass
AT-LOC-02	FR-LOC-02	Automated check finds no missing translation key in either language	Unit	Pass
AT-AUD-01	FR-AUD-01	Each action of the demo script produces exactly one matching audit entry	API	Pass
AT-AUD-02	FR-AUD-02	No endpoint updates or deletes audit entries; filters return the right entries	API	Pass
AT-MAT-01	FR-MAT-01	Ranking holds only eligible vacancies in descending fit; on the labelled set a relevant vacancy is in the top 5 for at least 80% of the profiles	API, Eval	Pass
AT-MAT-02	FR-MAT-02	Of two otherwise identical vacancies, the one whose required skills the student has scores higher	Unit	Pass
AT-MAT-03	FR-MAT-03	A vacancy whose minimum GPA is above the student's GPA is absent from the ranking and shows "GPA below minimum" when browsing	API	Pass

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Test	Requirement	Scenario and expected result	Level	Result
AT-MAT-04	FR-MAT-04	With the model off, the ranking is returned within 5 s and labelled basic	API, Unit	Pass
AT-MAT-05	FR-MAT-05	Matched and missing skills equal the set comparison of confirmed and required skills	Unit, API	Pass
AT-MAT-07	FR-MAT-07	The student can apply to the lowest-ranked eligible vacancy; ranking changes no application	API	Pass
AT-ATT-01	FR-ATT-01	"Full attendance" records attended days equal to scheduled days in one action; more days than scheduled is refused	API	Pass
AT-ATT-02	FR-ATT-02	An edit on day 8 after the week is refused until the coordinator unlocks the week; the unlock is in the audit log	API	Pass
AT-ATT-03	FR-ATT-03	The percentage equals attended ÷ scheduled days for seeded data; each role sees only its placements	API	Pass
AT-ATT-04	FR-ATT-04	A seeded unrecorded week appears on the company supervisor's dashboard on day 4, not on day 3	API	Pass
AT-EVL-01	FR-EVL-01	Submitting with an unrated criterion or an empty comment is refused	API	Pass
AT-EVL-02	FR-EVL-02	A score for a criterion of another cycle is refused; every evaluation of a test cycle uses that cycle's criteria	API	Pass
AT-EVL-03	FR-EVL-03	The student sees the periodic evaluation at once and the final evaluation only when the grade is Final	API	Pass
AT-EVL-04	FR-EVL-04	Editing a submitted evaluation is refused; the coordinator reopens it once with a reason, which is audited; a second reopening is refused	API	Pass
AT-WFL-02	FR-WFL-02	For seeded inputs the computed grade equals a hand calculation to two decimals	Unit, API	Pass
AT-WFL-03	FR-WFL-03	Submitting with any precondition unmet is refused with the list of missing items	API	Pass
AT-WFL-04	FR-WFL-04	Returning without a comment is refused; approval sets Final with a timestamp and completes the placement	API	Pass
AT-WFL-05	FR-WFL-05	Editing a Final grade is refused; reopening without a reason is refused; reopening is audited	API	Pass
AT-WFL-06	FR-WFL-06	The student's grade view is hidden in Draft and Submitted and shown when Final	API	Pass
AT-WFL-07	FR-WFL-07	A bulk approval of 10 grades creates 10 audit entries	API	Pass
AT-DSH-01	FR-DSH-01	For a seeded scenario, every dashboard count of every role equals the pending items in the database	API	Pass
AT-DSH-02	FR-DSH-02	Every cycle figure, overall and per company, equals the value computed from seeded data	API	Pass
AT-DSH-05	FR-DSH-05	Every platform figure equals the seeded value; no report, evaluation, or grade is reachable by the Platform Admin	API	Pass

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Test	Requirement	Scenario and expected result	Level	Result
AT-PER-02	NFR-PER-02	1,000 open vacancies: 95th percentile of 20 ranking requests ≤ 5 s, semantic and basic	API (timed)	Pass
AT-AI-02	NFR-AI-02	Each stored fit score and each application snapshot has the model version	API	Pass
AT-AI-03	NFR-AI-03	The held-out sets give their numbers in one run with the frozen settings; targets: at least 9 of 10 copied flagged and at most 1 of 10 originals flagged; a relevant vacancy in the top 5 for at least 80% of profiles	Eval	Matching met; similarity **not met**
AT-LOC-03	NFR-L10N-01, FR-LOC-02	The translation-key check passes with the new screens; each new screen checked in both languages	Unit, Demo	Pass
AT-SIM-01	FR-SIM-01	A report copied from a previous cycle's report of the same university is flagged; the same text stored at another university is not compared; earlier versions of the same report are not sources	API	Pass
AT-SIM-02	FR-SIM-02	Of 10 reports with 40% or more copied text, at least 9 are flagged with passages; at most 1 of 10 original reports is flagged. The academic supervisor sees score, cycle, week, and passages, not the author; the author's coordinator sees the author	API, Eval	Pass
AT-SIM-03	FR-SIM-03	A decision needs a comment and only a flagged check takes one; report status and grade do not change	API	Pass
AT-NOT-01	FR-NOT-01	Each FR-NOT-05 event creates one in-app notification and one email; the unread count drops when read	API	Pass
AT-NOT-02	FR-NOT-02	Reminders only for unsubmitted slots due within 24 hours, once each	API	Pass
AT-NOT-03	FR-NOT-03	Every seeded missing report gives a notice to the student and the academic supervisor within 1 hour	API	Pass
AT-NOT-04	FR-NOT-04	An escalation for each seeded overdue report, attendance week, and evaluation on day 4, none on day 3, never twice	API	Pass
AT-NOT-05	FR-NOT-05	Each listed event notifies the right user in a scripted test	API	Pass
AT-NOT-06	FR-NOT-06	A user with Arabic selected receives the Arabic email and notification text	API	Pass
AT-EXP-01	FR-EXP-01	Exported values equal stored values in every row of the Excel and PDF grade sheets; Arabic names are present in both	API, Unit	Pass
AT-EXP-02	FR-EXP-02	The term summary figures equal the dashboard figures	API	Pass
AT-EXP-03	FR-EXP-03	The training record matches the stored data of a seeded student; another supervisor is refused	API	Pass
AT-PWA-01	FR-PWA-01	The manifest and service worker meet the browser's install criteria; the app installs in desktop Chrome	Unit, Browser	Pass; the install itself on phones in M6

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Test	Requirement	Scenario and expected result	Level	Result
AT-PWA-02	FR-PWA-02	With a subscription the push is sent with only the title and a link; with none the email still goes out; a gone subscription is deleted	API	Pass; a real push on a device in M6
AT-PWA-03	FR-PWA-03 (CR-10)	Offline: a page opened before shows its saved copy with the time; a page never opened shows that it is not saved; a report draft typed offline is saved after the connection returns; submit buttons stay disabled; after sign-out nothing is left in session storage and no draft is left	E2E, Browser	Pass
AT-AUD-03	FR-AUD-03	The history shown for a report, an evaluation, and a grade equals their audit entries; other users are refused	API	Pass
AT-ORG-07	FR-ORG-07	After a transfer the previous holder loses the rights and the new one has them; a reason is required; the transfer is in the audit log; both are notified	API	Pass
AT-PER-04	NFR-PER-04	With 10,000 stored reports the similarity result is ready within 2 minutes	API (timed)	Pass
AT-PER-05	NFR-PER-05	100-student grade sheet in Excel and PDF in under 60 s each	API (timed)	Pass
AT-REL-03	NFR-REL-03	Running each job twice sends nothing twice; after a simulated downtime the next run catches up	API	Pass
AT-L10N-03	NFR-L10N-03	The exported PDF embeds the Arabic font and its text holds the shaped Arabic names; checked visually	Unit, Browser	Pass
AT-CER-01	FR-CER-01	A certificate cannot be issued for a grade that is not Final; issued for a Final grade, it has a 26-character code, and its PDF holds the code and a QR code that opens `/verify/<code>`; a second valid certificate for the same placement is refused	API, Unit	Pass
AT-CER-02	FR-CER-02	Without login, a valid code shows Valid with the listed fields and no grade; a revoked code shows Revoked; an altered code shows Not found	API, E2E	Pass
AT-CER-03	FR-CER-03	Revoking needs a reason and is audited; after it the page shows Revoked, and a new certificate can be issued	API	Pass
AT-EVL-05	FR-EVL-05	The student evaluates the company once, only after the training end date or a Final grade; the coordinator sees it and the averages per company; a company supervisor and a Company Admin are refused	API	Pass
AT-DSH-03	FR-DSH-03	With seeded answers, the job-offer rate equals Yes answers ÷ survey answers, per cycle and per company; a student who skipped is not counted	API	Pass
AT-DSH-04	FR-DSH-04	A company with 5 final evaluations averaging 4.8 against a cycle mean of 3.6 raises the alert; a company with 4 evaluations does not; the alert changes nothing else	API	Pass

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Test	Requirement	Scenario and expected result	Level	Result
AT-PRV-04	NFR-PRV-04	Before the retention period ends, anonymizing is refused; after it, no name, email, student number, CV, report text, or certificate file of the cycle can be retrieved, students still in another active cycle are skipped, and the cycle statistics are unchanged	API	Pass
AT-SEC-05	NFR-SEC-05	ZAP baseline with no high-risk finding; pip-audit and npm audit with no critical finding	Scan	Pass
AT-PER-01	NFR-PER-01	50 users: 95th percentile of ordinary requests under 2 s	Load	Pass on the local stack
AT-PRT-01	NFR-PRT-01, NFR-USA-03	The browser tests pass on the three engines and at 360, 768, 1366, and 1920 px without horizontal scrolling	E2E	Pass; Firefox in CI, real phones in the UAT
AT-USA-04	NFR-USA-04	The accessibility check finds no contrast failure on the main pages; the main tasks can be done with the keyboard	E2E, Browser	Pass

11.3 References

1. IEEE Std 829-1998, *IEEE Standard for Software Test Documentation*, IEEE Computer Society, 1998.

2. IEEE Std 1016-1998, *IEEE Recommended Practice for Software Design Descriptions*, IEEE Computer Society, 1998.

3. I. Sommerville, *Software Engineering*, 10th ed., Pearson, 2016.

4. Group 3, *InternSync – Software Requirements Specification*, version 1.6, Oct 28, 2026.

5. Group 3, *InternSync – Software Design Description*, version 1.0, Nov 1, 2026.

6. Group 3, *InternSync – Test Reports of Increments 1 to 5* and *Key Results*, Oct 24 – 29, 2026.

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Evaluation

To be completed by the instructor or supervisor.

7. Project STS Report (12 points / 2 marks) – Week 15

Section	Excellent (Full Points)	Good (75%)	Fair (50%)	Poor (25% or below)	Points
1. Introduction (1 pt)	Purpose, scope, and references clearly stated.	Covers most but vague.	Minimal explanation.	Missing.	/1
2. Test Items (1 pt)	All modules/features listed clearly.	Most features listed.	Few features listed.	Missing.	/1
3. Features to Be Tested (1.5 pts)	Comprehensive list mapped to SRS/SDS.	Most key features covered.	Few features listed; weak traceability.	Missing.	/1.5
4. Features Not to Be Tested (0.5 pt)	Clearly explains out-of-scope features.	Partially explains.	Vague or incomplete.	Missing.	/ 0.5
5. Test Approach (1.5 pts)	Clear strategy (levels, tools, methods).	Mostly clear, some gaps.	Very limited explanation.	Missing.	/1.5
6. Test Cases (3 pts)	Well-structured test case table with inputs, expected/actual outputs, pass/fail. Covers positive & negative tests.	Several test cases included, some incomplete.	Few test cases, weak structure.	Missing.	/3
7. Test Data (1 pt)	Relevant datasets defined (valid & invalid).	Some datasets defined.	Minimal test data.	Missing.	/1
8. Test Environment (1 pt)	Hardware/software clearly described.	Partial description.	Very minimal.	Missing.	/1
9. Responsibilities & Schedule (1 pt)	Clear team roles & testing timeline.	Some roles/schedule mentioned.	Very vague or incomplete.	Missing.	/1
10. Deliverables &	All deliverables listed; glossary	Most listed,	Minimal	Missing.	/1.5

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Section	Excellent (Full Points)	Good (75%)	Fair (50%)	Poor (25% or below)	Points
Appendices (1.5 pts)	& references included.	minor gaps.	mention of deliverables.		

Note: If a student is listed as a project member at the beginning of the STS but their **contributions are not reflected** (e.g., no assigned test cases, no role in responsibilities/schedule), that student will **not be awarded the same score** as contributing members.